SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS BASED ON FIELD VALUE

1. Introduction

ServiceNow is a cloud-based platform used by organizations to manage their daily activities, services, and business processes. It provides different security features to protect important information from unauthorized users.

This project, **Script-Controlled ACL – Restrict Record Access Based on Field Value**, is developed using the ServiceNow platform. The main purpose of this project is to control access to records based on user roles and specific field values.

In organizations, all employees should not have access to every record. For example, an employee should only view their own records, while a manager may view records belonging to their department. An HR manager may have permission to view all employee records.

To achieve this, ServiceNow uses Access Control Lists (ACLs). JavaScript scripts are added to ACLs to check whether a user has permission to access a particular record.

This project demonstrates how to provide secure access to records using conditions such as employee name, assigned user, department, and user role.

2. Project Overview

The project creates an Employee Management application in ServiceNow. It contains employee details such as employee ID, employee name, department, assigned user, and salary.

Using scripted ACLs, the system checks the logged-in user and record information before allowing access.

If the user satisfies the access conditions, permission is granted. Otherwise, access is denied.

3. Objectives

The main objectives of this project are:

1. To provide secure access to employee records.
2. To restrict record access based on field values.
3. To use JavaScript scripts in ServiceNow ACLs.

4. To allow employees to access only their own records.
5. To allow managers to access records based on their departments.
6. To protect sensitive information such as employee salaries.
7. To prevent unauthorized users from accessing records.

4. Scope of the Project

This project focuses on implementing record-level and field-level security in ServiceNow.

The application is designed for employee information management. Different users are provided with different access permissions based on their responsibilities.

The project includes:

- Creating a custom Employee Details table.
- Creating employee records.
- Creating user roles.
- Configuring Read and Write ACLs.
- Writing JavaScript access validation scripts.
- Testing access permissions using different users.

5. Need for the Project

In many organizations, employee information contains sensitive details. If all users can access every record, it may lead to unauthorized access and data security problems.

Therefore, a security mechanism is needed to control who can view or modify information.

This project solves the problem by checking user roles and field values before providing access to records.

6. Technologies Used

The following technologies are used to develop this project.

1. ServiceNow

ServiceNow is the main platform used to create the Employee Management application, tables, users, and security rules.

2. JavaScript

JavaScript is used to write scripts inside ACLs. These scripts check user roles and record field values to decide whether access should be granted or denied.

3. Access Control Lists (ACLs)

ACLs are security rules that control access to ServiceNow tables and fields. They are used to manage Read, Write, Create, and Delete permissions.

4. Glide System

Glide System is used to get information about the currently logged-in user and check user roles.

5. ServiceNow Tables

Tables are used to store employee information such as employee name, employee ID, department, and salary.

6. User Roles

Roles are assigned to users to provide different levels of access based on their responsibilities.

7. Hardware and Software Requirements

Hardware Requirements:

- Computer or laptop.
- Internet connection.
- Minimum 4 GB RAM.

Software Requirements:

- Web browser.
- ServiceNow Developer Instance.
- JavaScript.
- Internet connection.

8. System Design

The project works by checking the user's identity and record details whenever a user tries to access an employee record.

The basic workflow is:

1. User logs into ServiceNow.
2. User opens the Employee Details table.

3. User selects a record.
4. The ACL script executes.
5. The script checks the user's role and record field values.
6. If the conditions match, access is granted.
7. Otherwise, access is denied.

9. Project Modules

Module 1: User Management

This module is used to create users and assign roles such as Employee, IT Manager, and HR Manager.

Module 2: Employee Management

This module is used to create and maintain employee records. It stores employee details such as name, ID, department, assigned user, and salary.

Module 3: ACL Configuration

This module is used to create Access Control Lists for controlling Read and Write permissions.

Module 4: Script Validation

This module uses JavaScript to check the user's role and record field values before granting access.

Module 5: Testing

This module checks whether authorized users can access records and unauthorized users are denied access.

10. Implementation Steps

The project is implemented using the following steps in ServiceNow.

Step 1: Create a Developer Instance

Register on the ServiceNow Developer Portal and create a Personal Developer Instance (PDI).

Step 2: Create an Employee Details Table

Navigate to System Definition → Tables and create a new table named Employee Details.

Step 3: Create Fields

Add the following fields:

- Employee Name
- Employee ID
- Assigned To
- Department
- Salary
- Status

Step 4: Insert Employee Records

Create sample employee records with different departments and assigned users.

Step 5: Create User Roles

Create three roles:

- Employee
- IT Manager
- HR Manager

Step 6: Assign Roles

Assign the required roles to the corresponding test users.

Step 7: Create ACL Rules

Navigate to System Security → Access Control (ACL) and create the required Read and Write ACLs.

Step 8: Add JavaScript Scripts

Add scripts to check the logged-in user, role, and field values.

Step 9: Test the Application

Use different user accounts to check whether the access restrictions work correctly.

11. Script-Controlled ACL

The main part of this project is the JavaScript code used inside the ACL.

The Read ACL checks whether a user is allowed to view a particular employee record.

Example: Read ACL Script

Java script

(function() {

```
  // HR Manager can read all records
  if (gs.hasRole('hr_manager')) {
    answer = true;
  }
  // Employee can read their own record
  else if (
    current.u_assigned_to.toString() ==
    gs.getUserID()
  ) {
    answer = true;
  }
  // IT Manager can read records
  // from their department
  else if (
    gs.hasRole('it_manager') &&
    current.u_department.toString() ==
    gs.getUser().getDepartmentID()
  ) {
    answer = true;
  }

  // Deny access to other users
  else {
    answer = false;
  }

})();
```

12. Write ACL Script

The Write ACL controls who can update employee records.

Javascript

```
(function() {
  // HR Manager can update all records
  if (gs.hasRole('hr_manager')) {
    answer = true;
  }
  // Employee can update their own record
  else if (
    current.u_assigned_to.toString() ==
    gs.getUserID()
  ) {
    answer = true;
  }
  // Deny other users
  else {
    answer = false;
  }
})();
```

13. Salary Field Protection

A separate field-level Read ACL is created for the Salary field.

javascript

```
answer = gs.hasRole('hr_manager');
```

This allows only users with the HR Manager role to read salary information, provided no other ACL grants access.

14. Testing

Testing is performed to verify whether the ACL scripts are working correctly.

Different users are used to check the access permissions.

15. Test Cases

User	Read Access	Write Access
Employee	Own records	Own records
IT Manager	IT department records	Denied
HR Manager	All records	All records
Unauthorized User	Denied	Denied

Salary field access is restricted to HR Managers.

16. Expected Results

After implementing the ACL rules, the system should provide the following results:

1. Employees can view their own records.
2. Employees can update their own records.
3. IT Managers can view records belonging to their department.
4. HR Managers can view and update all employee records.
5. Unauthorized users cannot access restricted records.

17. Advantages

- Improves data security.
- Prevents unauthorized record access.
- Protects sensitive employee information.
- Provides flexible access permissions.

18. Conclusion

The Script-Controlled ACL project is developed using the ServiceNow platform to provide secure access to employee records.

In this project, Access Control Lists and JavaScript scripts are used to restrict access based on user roles and record field values.

Employees can access their own records, managers can access records based on their departments, and HR Managers can access all employee information.

The project helps to improve data security and prevent unauthorized access to important information.